

Adapting a Large Multilingual Model to Cornish: A Case Study in Low-Resource Machine Translation

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Abstract

Recent work in low-resource NMT has focused on transfer learning, specifically realised through the use of large multilingual translation models. This project investigates whether a multilingual neural machine translation model can be effectively adapted to translate to Cornish, a severely low-resource Celtic language with limited parallel data. Using approximately 27,000 English-Cornish sentence pairs from Korpus Kernewek and Tatoeba, two variants of the multilingual NLLB-200 model were fine-tuned: one in which the newly added Cornish language token was initialised randomly, and one in which it was initialised using the Welsh embedding to encourage transfer from a closely related Celtic language. Both models achieved strong performance: up to 51.22 BLEU and 69.98 ChrF2++. However, results showed that the Welsh-initialised model does not yield a statistically significant improvement over the randomly initialised one in performance or rate of convergence. In summary, this study highlights the potential of adapting large multilingual models to low-resource languages and provides a basis for future work on dedicated Cornish MT systems.

1 Introduction

Cornish is a Celtic language spoken in Cornwall, a region located at the southwestern tip of England. Its history as a minority language is quite distinctive, as it has seen a revival after being classified as extinct in the late 18th to early 19th centuries. According to Davies-Deacon (2022), much of the more recent work on Cornish has focused around sociolinguistic and political aspects, particularly the events that lead to its decline and the revitalisation efforts that followed. Based on a 2011 census, 562 people answered with Cornish when asked "What is your main language?", a number

that, according to Lowe (2018), is an underestimate, as the question does not accurately capture the language abilities of the population. In any case, Cornish remains a small and resource-poor language community.

Despite these challenges, Cornish has begun to appear in the context of computational linguistics. Recent work by Prys (2020), supported by Akademi Kernewek (Cornish Academy), led to the development of an official online Cornish dictionary by adapting Welsh tools to accommodate the needs of Cornish. Such efforts indicate that there is a demand for digital language tools within the Cornish community, and that learners and users find value in computational support. However, Cornish remains a severely low-resource language with little parallel data, and no Neural Machine Translation (NMT) systems comparable to those developed for other Celtic languages.

Current research on low-resource NMT focuses on the use of large multilingual pre-trained models (mBART, NLLB) and the use of synthetic data created through back-translation of monolingual data (Haddow et al., 2022; Sant et al., 2024; Dabre et al., 2023). Prior work (Galiano Jimenez et al., 2024) shows that typologically similar auxiliary languages and multilingual models can improve low-resource translation quality. Many contributions to the WMT24 Shared task on translating into low-resource languages of Spain (summarised in Sánchez-Martínez et al., 2024) report drastic improvements in all evaluation metrics when fine-tuning multilingual models such as NLLB-200, compared to the previous state-of-the-art. Yet no work has examined NMT for Cornish specifically.

The No Language Left Behind (NLLB) project by Meta AI produced a multilingual many-to-many translation model capable of translating between 200+ languages (over 40,000 language pairs), with its primary focus being low-resource languages (NLLB-Team et al., 2022). While Cor-

nish was not included in the training corpus, we are still able to leverage its power because of the several related Celtic languages that are represented (Welsh, Irish, and Scottish Gaelic), as well as cross-linguistic generalisations the model has learned.

NLLB relies on special tokens to signal the source and target languages during translation. The model prepends the source language token to the source sequence and produces the target language token first in the output sequence. When adding a language token to NLLB models, it is common practice to initialise its embeddings on related languages that are already represented in the model (Galiano Jimenez et al., 2024; Mutal and Ormaechea, 2024). However, that choice is rarely rationalised through direct comparative evaluation. Initialising the Cornish language token using the embedding of a closely related Celtic language, such as Welsh, will theoretically position Cornish closer to other Celtic languages in the model’s vector space, potentially allowing the model to benefit from shared orthographic and typological properties.

Based on this motivation, the following hypothesis is proposed:

- **H1:** Initialising the Cornish language token embedding using Welsh, a typologically related language, will lead to improved translation performance and faster convergence compared to random initialisation.

The purpose of this study is to assess whether a multilingual model such as NLLB-200 can be effectively fine-tuned for English-to-Cornish translation and to examine whether different embedding initialisation strategies influence performance in this low-resource setting. By doing so, this study aims to provide a basis for future work on Cornish MT tools and to support ongoing efforts in the language’s preservation.

This study addresses the following research questions:

- **RQ1:** To what extent can NLLB-200 be fine-tuned to produce effective English–Cornish translations in a severely low-resource setting?
- **RQ2:** How does the choice of embedding initialisation (random vs. Welsh-based) affect translation performance?

2 Data

Parallel data in Cornish, as with all low-resource languages, is scarce by definition. This is why it is crucial that the data used be of the highest quality, since models trained on little data are especially prone to overfitting to noise. Taking that into consideration, two corpora were chosen for the purposes of this project - Korpus Kernewek and Tatoeba.

Korpus Kernewek. Akademi Kernewek released part of their corpus dataset under the CC0 1.0 Universal license (also available online¹), and is by far the largest English-Cornish parallel corpus available, containing 24,492 sentence pairs (version 2025.09.17). These translations are provided by Gonis Treylya, the Cornish Language Translation Service, and abide by the standard written form (SWF) rules for the Cornish language.

Tatoeba. Publicly available on OPUS (Tiedemann, 2012), the Tatoeba dataset (version 2023.04.12) consists of 4,198 English-Cornish sentence pairs derived from the online database of the same name². This data is released under the CC-BY 2.0 FR licence.

2.1 Data pre-processing

The Tatoeba dataset sometimes provides alternative translations for the same source sentence, meaning that some source sentences may appear twice or, in some cases, up to seven times. This may be problematic further down the pipeline, as there is a high potential for overlap between the train and test sets, which in turn could inflate evaluation scores. Therefore, as a first step, only the first occurrence of each source sentence is kept, while every subsequent occurrence is discarded.

After merging the two datasets, exact duplicate sentence pairs, where both the source and target texts are identical, are removed to further mitigate data leakage.

Finally, the data is divided using a 95:5 train/test split, maintaining the original ratio between the two datasets. The choice of not using a dev set was made in order to limit the amount of data that is held out from the already limited training data.

¹<https://www.akademikernewek.org.uk/corpus/>

²<https://tatoeba.org>

	train	test	total
Original	-	-	28,690
Discarded	-	-	1,507
Final	25,824	1,359	27,183

Table 1: Number of sentence pairs in the combined dataset before and after data pre-processing.

3 Method

3.1 Models

This project compares two variants of the NLLB-200-distilled-600M model. As already mentioned, NLLB relies on special tokens to signal the source and target languages during translation. Since Cornish is not natively supported, a new special token ("cor_Latn") was added to represent it, meaning that the model’s embedding matrix needed to be expanded accordingly.

In the first variant, the embedding of the newly added language token is initialised randomly, whereas in the second, the embedding is copied over from Welsh, one of the most closely related Celtic languages that is available in the model’s repertoire. Finally, the tokenizer was not trained on the Cornish data, since NLLB uses the language-agnostic SentencePiece tokenizer that has already been trained on related languages with similar orthographic patterns.

3.2 Configuration

Both models are fine-tuned on the combined English-to-Cornish train set on a T4 GPU on Google Colab. The models were trained using the Adafactor optimizer, with a learning rate of 10^{-4} , a clip threshold of 1.0 and a weight decay of 10^{-3} . A constant learning rate scheduler was used with 500 warm-up steps. Due to computational limitations, a batch size of 8 was used, and the maximum length was set to 128 tokens per sentence. Both models were trained for a total of 8 epochs, which was chosen empirically based on the BLEU score after evaluating on the test set for epochs 1-10.

In the decoding step, translations were generated using beam search with a beam size of 5. To mitigate the tendency of encoder-decoder models such as NLLB to produce repetitive sequences of tokens, decoding was performed with a no-repeat-ngram constraint of 3, preventing the model from generating any three-token sequence more than once within a translation.

3.3 Reproducibility

All experiments were conducted with a fixed seed to promote reproducibility. The seed was applied to model weight initialisation, data splitting and shuffling, and other stochastic components of the fine-tuning process. The exact code and data used are publicly available at <https://github.com/skamateros/cornish-nmt>.

3.4 Metrics

Two evaluation metrics were chosen to assess the models’ translation quality. BLEU (Papineni et al., 2002), the most widely adopted metric in machine translation, offers a familiar method of comparison across systems. Additionally, ChrF2++ was used, as it has been shown to align more closely with human judgement (Popović, 2015, 2016).

Both metrics were calculated at the corpus level using the sacreBLEU Python library along with default settings. For ChrF2++, word n-grams up to order 2 were enabled by setting `word_order=2` when initialising the `sacrebleu.CHRF` class. Corpus-level evaluation refers to computing BLEU and ChrF2++ once over the entire test set, rather than averaging scores computed for individual sentences, and is standard practice in machine translation.

4 Results

The training loss for both models was collected at each epoch’s mid-point and end-point, the values of which can be seen in Table 2. The loss for the two models is nearly identical, indicating that the choice of language token initialisation does not noticeably affect the rate at which the model learns.

After producing translations of the test set using the aforementioned parameters, BLEU and ChrF2++ scores were calculated using the sacreBLEU Python library and the parameters that are specified in section 3.4, the results of which can be seen in Table 3. The two models achieve almost identical performance, with the BLEU score slightly favouring the Welsh-init model, and ChrF2++ slightly favouring the Random-init model.

To assess whether these small differences were statistically significant, a paired bootstrap test with 4,000 samples was performed on the BLEU scores. Paired bootstrap tests are naturally suited

Epoch	Random-init	Welsh-init
0.5	3.0494	2.8410
1.0	1.1733	1.1852
1.5	0.8462	0.8404
2.0	0.6607	0.6699
2.5	0.5436	0.5607
3.0	0.4659	0.4629
3.5	0.3970	0.4097
4.0	0.3381	0.3469
4.5	0.2981	0.3018
5.0	0.2640	0.2670
5.5	0.2308	0.2351
6.0	0.2070	0.2067
6.5	0.1805	0.1839
7.0	0.1677	0.1620
7.5	0.1482	0.1481
8.0	0.1328	0.1336

Table 2: Training loss across epochs for random and Welsh-based initialisation models.

	Random-init	Welsh-init
BLEU	50.99	51.22
ChrF2++	69.98	69.72

Table 3: The two models’ BLEU and ChrF2++ scores on the test set. Highest scores are presented in bold.

for this setting, as BLEU is a corpus-level metric, and both models are evaluated on the same test sets. As shown in Table 4, the confidence interval includes zero and the estimated probability that the Welsh-init model outperforms the Random-init model is 0.68, indicating that the difference is not statistically significant.

$\delta(x)$	$p(B > A)$	95% CI
0.2246	0.6755	[-0.8063, 1.2360]

Table 4: Paired bootstrap test on BLEU scores with 4,000 samples (A: Random-init, B: Welsh-init; $\delta(x) = \text{BLEU}(B) - \text{BLEU}(A)$; $p(B > A)$: probability that B outperforms A; CI: 95% confidence interval).

5 Discussion

This study demonstrates that the NLLB-200 multilingual model can be effectively fine-tuned for English-to-Cornish translation, achieving BLEU scores over 50 and ChrF2++ scores approaching 70. These results are encouraging given the scarcity of parallel data available, and provide further evidence that pre-training multilingual models is a viable method for low-resource MT.

Regarding RQ2 and the two initialisation strategies, the two models produce nearly identical performance. While the hypothesis that initialising the Cornish token within the Celtic space would benefit the model or provide some gain in efficiency seems logical, the statistical analysis shows that there is no significant advantage. The training loss (Table 2) between the two models further support these results, as both models converge at similar rates. Therefore, H1 is rejected. Several factors could explain these findings. First, the model’s extensive pre-training on related languages such as Welsh, Irish, and Scottish Gaelic may catalyse the fine-tuning process at a positively disproportionate rate, making the effects of Welsh initialisation negligible. Another possible explanation is that despite the lexical and orthographical similarities between Cornish and Welsh, the benefit is imperceptible for this particular configuration.

While a comprehensive qualitative analysis would only be possible with the help of a native speaker, a preliminary examination of the model’s outputs reveals some error patterns. The following judgments are based on corpus searches from Korpus Kernewek. Table 5 presents a small set of examples that illustrate common error types.

First, not all mismatches represent errors; in some cases, the model produces acceptable synonyms or paraphrases that are penalised by automatic evaluation metrics. In sentence 1, the model translates the phrase *I love* with the construction *My a gar*, which, based on corpus searches, is a common alternative to the expected *Y karav*.

A clear weakness, on the other hand, is presented in sentence 2. In this case, the model’s mistake involves both the gendering of the cardinal number, as well as a consonant mutation. It seems that the model inflects the number in the masculine, while the noun *pluven* is feminine, and therefore collocates with *teyr*. Moreover, both masculine and feminine forms of the number *three* cause a consonant mutation in the following word, turning *p* in p-initial words into *f*, which the model fails to do. Sentence 3, however, demonstrates that the model can apply this mutation correctly in a similar context. Context-dependent mistakes such as these are unsurprising given the limited amount of data and would likely decrease if the model were trained on a larger dataset.

Finally, the model is inconsistent with regard to the use of apostrophes in contracted words (see

#	Source	Target	Output
1	I love Cornwall	Y karav Kernow	My a gar Kernow
2	You have three pens.	Yma teyr fluven dhis.	Yma tri pluven dhis.
3	My pen is red.	Ow fluven yw rudh.	Ow fluven yw rudh.
4	Do we study in the college?	A wren ni studhya y'n kolji?	A wren ni studhya yn kolji?
5	Where is my car?	Ple'ma ow harr-tan?	Plema ow harr-tan?

Table 5: Sample translations from the test set, generated by the Welsh-initialised model. The source and target columns correspond to the values of the test set, while output contains the model’s translations.

sentences 4 and 5), which it occasionally omits. Although this represents a minority of cases, it still occurs enough to pose an issue, as there is no indication in the bibliography that the apostrophes are optional.

6 Conclusion

This study investigated the adaptation of a large multilingual model, NLLB-200, for English-to-Cornish translation in a low-resource setting. Two embedding initialisation strategies for the Cornish language token were compared, random and Welsh-based.

The results show that both models achieve strong performance, with BLEU scores up to 51.22 and ChrF2++ scores up to 69.9, and that NLLB-200 adapts well to English–Cornish translation despite the limited data available. However, the choice of token initialisation had no significant effect on performance or convergence rate in the current context. The error patterns observed suggest areas which could benefit from more data or further targeted fine-tuning. Finally, a good next step would be a more robust qualitative analysis conducted by native or proficient speakers, which would provide a clearer picture of the model’s weaknesses and guide future work.

7 Limitations

As is natural in low-resource settings, the limited size of the available corpus introduces a domain bias, and it remains unclear how well the model generalises to other domains. In addition, due to time and compute constraints, no English-Cornish baseline model was trained directly on the parallel data. Such a baseline would have provided a more definitive indication of the gains that come with using multilingual models relative to training a model from scratch, and could be addressed in future work.

8 Ethical Considerations

All datasets used in this study were handled in accordance with their respective licenses. As most of the training data come from a single source, there is a potential bias towards the stylistic and orthographic choices of the translators, which may not reflect the full range of Cornish usage. Additionally, given the low number of total speakers, and considering that many potential users are learners, it is crucial to acknowledge the limitations of the current model, as any low-quality and erroneous translations may influence language practices within the community. Therefore, any translation system should be thoroughly evaluated by proficient speakers before deployment.

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